

Q3LOCK Modular Cutoff and Unitary Resummation Route Split

TECT verification-first repository
claim C6-SPACETIME-SIGNATURE

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1 Purpose and scope

This note records **EXP-000798** and strengthens reusable result **R-167** to v1.2 without allocating a new result number. It is an additive route correction: the v1.0–v1.1 weighted-energy and cubic graph theorems remain unchanged.

The fixed-energy-power, coefficientwise first-passage target left by v1.1 is false. Exact star and repeated-bond commutators accumulate one coordinate per raw branch or repeat. The same star nevertheless resums over all orders to an exact unitary phase. A unique-path tree has an exact sequential-activation Duhamel formula, but a square leaves an alternate-path remainder.

The fixed-temperature alternative also sharpens. An exact arithmetic versus logarithmic mean inequality upgrades Duhamel convergence plus one bounded modular derivative to a two-sided GNS topology. Existing Euclidean exponential moments make a coordinate-only bond cutoff and its first modular derivative exponentially small. An exact two-level fixture shows why an arbitrary bounded multiplier is still insufficient.

No common thermodynamic automorphism, common-alpha KMS identification, algebraic ground states, broken-sector GNS or physical mass gap, continuum removal, physical empty-space comparison, C6, CP1, Sector A or Pre-A closure is claimed.

2 Provenance identifiers

- Result: [PA-CP1-ST8-Q3LOCK-SECOND-WEIGHTED-ENERGY-MOMENT-AND-COMMON-ALPHA-CAUCHY-GATE-SPLIT](#), v1.2, R-167.
- Exploration: [EXP-000798](#).
- Historical gate: [PA-CP1-ST8-Q3LOCK-FIRST-PASSAGE-BACKBONE-REAL-TIME-PRODUCT-AND-ENERGY-TAIL-CLOSURE](#).
- Historical gate: [PA-CP1-ST8-Q3LOCK-FIFTH-ENERGY-MOMENT-AND-MODULAR-CUTOFF-LOCALITY](#).
- Active gate: [PA-CP1-ST8-Q3LOCK-ALL-BOND-UNITARY-TROTTER-GRAPH-LIPSCHITZ-AND-COMMON-ALPHA-CLOSURE](#).
- Active gate: [PA-CP1-ST8-Q3LOCK-PROJECTED-DUHAMEL-MODULAR-C1-MULTIPLIER-LOCALITY](#).
- Negative: [NG-2026-08-10-PRE-A-ST8-Q3LOCK-FIXED-S-COEFFICIENTWISE-FIRST-PASSAGE-BRANCH-RESPONSE](#).
- Negative: [NG-2026-08-10-PRE-A-ST8-Q3LOCK-STATIC-MODULAR-TAIL-ARBITRARY-BOUNDED-MULTIPLIER](#).

3 Fixed-order star obstruction and exact resummation

For one component, let

$$V_j = -cq_0q_j, \quad W_0(a) = e^{-iap_0/\hbar}.$$

The Weyl relation and commutativity of all bond multipliers give

$$\text{ad}_{V_m} \cdots \text{ad}_{V_1} W_0(a) = (-ca)^m \left(\prod_{j=1}^m q_j \right) W_0(a), \quad (3.1)$$

with the same sign for every ordering. One repeated edge gives

$$\text{ad}_V^n W_0(a) = (-ca)^n q_y^n W_0(a). \quad (3.2)$$

On translated smooth product bumps,

$$\|A^s \psi_R\| = O(R^{4s}), \quad \left\| \prod_{j=1}^m q_j W_0(a) \psi_R \right\| \gtrsim R^m. \quad (3.3)$$

Thus both one-sided graph operators fail whenever $m > 4s$. The first failures are $m = 3$ at $s = 1/2$ and $m = 4$ at $s = 3/4$, both inside a degree-six lattice star. This rejects the literal coefficientwise target, not the dynamics.

Indeed the complete star subflow is

$$\text{Ad} \left(e^{it \sum_j V_j/\hbar} \right) W_0(a) = \exp \left[-\frac{icat}{\hbar} \sum_j q_j \right] W_0(a), \quad (3.4)$$

an exact unitary. Cancellation occurs across all perturbative orders.

4 Tree theorem and loop boundary

On a tree, let $e_1, \dots, e_d$ be the unique path between the observable supports. Absorb every off-path component into H_0 and put

$$H_k = H_0 + \sum_{r \leq k} V_{e_r}, \quad \mathcal{U}_k(t) = \text{Ad}(e^{itH_k/\hbar}), \quad \mathcal{D}_r = \frac{i}{\hbar} \text{ad}_{V_{e_r}}.$$

Successive Duhamel identities give

$$[\mathcal{U}_d(t)A, B] = \int_{0 < t_1 < \dots < t_d < t} [\mathcal{U}_d(t - t_d) \mathcal{D}_d \mathcal{U}_{d-1}(t_d - t_{d-1}) \cdots \cdots \mathcal{D}_1 \mathcal{U}_0(t_1)A, B], dt. \quad (4.1)$$

Each path edge appears once; all side branches and post-activation repeats are inside exact unitaries. On the square $0 - 1 - 2 - 3 - 0$, however, deleting the chosen backbone $0 - 1 - 2$ leaves $0 - 3 - 2$. The base propagator can already connect the endpoints. A cubic lattice therefore needs a cut/forest organization or a global bond resummation.

5 All-bond Trotter successor

Write

$$H = \sum_x h_x - c \sum_{\langle xy \rangle} q_x \cdot q_y. \quad (5.1)$$

Every cross bond commutes with every other one, including bonds sharing a site. The complete bond kick factorizes exactly and acts by

$$q_x \mapsto q_x, \quad p_x \mapsto p_x + \delta c \sum_{y \sim x} q_y. \quad (5.2)$$

The onsite flow is tensor-local. If a two-sided graph-Lipschitz core can be found with the one-step recurrence

$$L'_x \leq (1 + C\delta)L_x + J\delta \sum_{y \sim x} L_y, \quad (5.3)$$

then the exactly d -transfer coefficient after N steps is

$$\binom{N}{d} (Jt/N)^d (1 + Ct/N)^{N-d} \longrightarrow e^{Ct} \frac{(Jt)^d}{d!}. \quad (5.4)$$

The graph-Lipschitz core, uniform Trotter convergence, invariant Hamiltonian algebra, and KMS identification are open. Equation (5.4) is a conditional reduction, not a completed locality theorem.

6 Duhamel-to-GNS modular theorem

For a faithful Gibbs density ρ in one fixed finite-volume type-I representation, with X and $[\log \rho, X]$ in the Duhamel form domain, define

$$\|X\|_D^2 = \int_0^1 \text{Tr}(\rho^{1-s} X^* \rho^s X), ds, \quad \|X\|_{\#}^2 = \frac{1}{2} \text{Tr} \rho(X^* X + X X^*). \quad (6.1)$$

The respective matrix weights are the logarithmic and arithmetic means. If $u = |\log p_m - \log p_n|$, then

$$\frac{A(p_m, p_n)}{L(p_m, p_n)} = \frac{u}{2} \coth \frac{u}{2} \leq 1 + \frac{u}{2}. \quad (6.2)$$

Cauchy–Schwarz gives the exact inequality

$$\boxed{\|X\|_{\#}^2 \leq \|X\|_D^2 + \frac{1}{2} \|X\|_D \|[\log \rho, X]\|_D.} \quad (6.3)$$

With $\delta = (i/\hbar)[H, \cdot]$, $[\log \rho, X] = i\beta\hbar\delta X$. Hence Duhamel convergence and a uniform first modular-derivative bound imply two-sided GNS convergence. On a uniformly operator-bounded set this yields strong-star convergence in one fixed faithful type-I representation. Faithfulness (or support restriction) is essential; this matrix proof is not a general W-star modular-operator theorem, and it does not create a common representation across volumes.

For a modular-analytic multiplier B , put

$$M_0(B) = \sup_{|t| \leq 1/2} \|\rho^t B \rho^{-t}\|, \quad M_1(B) = M_0([\log \rho, B]). \quad (6.4)$$

Then

$$\|[X, B]\|_D \leq 2M_0\|X\|_D, \quad (6.5)$$

$$\|[\log \rho, [X, B]]\|_D \leq 2M_0\|[\log \rho, X]\|_D + 2M_1\|X\|_D. \quad (6.6)$$

These are exact sufficient multiplier conditions. Uniform M_0, M_1 for the structured truncated evolution remain open.

7 Why an arbitrary bounded multiplier still fails

Let

$$H_n = \text{diag}(0, n), \quad \rho_n = \frac{\text{diag}(1, e^{-\beta n})}{1 + e^{-\beta n}},$$

$$W_n = e^{\beta n/4}|1\rangle\langle 1|, \quad B_n = |1\rangle\langle 0|.$$

Then $[H_n, W_n] = 0$ and

$$\varphi_n(W_n^2) = \frac{e^{-\beta n/2}}{1 + e^{-\beta n}} \rightarrow 0,$$

but

$$\|[W_n, B_n]\|_D^2 = e^{\beta n/2} \frac{1 - e^{-\beta n}}{(1 + e^{-\beta n})\beta n} \rightarrow \infty. \quad (7.1)$$

Thus static tail smallness and even a zero modular derivative do not form an ideal under arbitrary contractions. The projected or structured multiplier condition is load-bearing.

8 Coordinate-only cutoff

The exact Q3LOCK hypothesis map in EXP-000781 imports the infinite-volume tempered-DLR one-site exponential path estimate of Kozitsky–Pasurek. For the finite periodic-volume bridge define

$$A_x = \lambda_\sigma \|\omega_x\|_{C^\sigma}^2 + \kappa \|\omega_x\|_2^2. \quad (8.1)$$

Finite-volume confinement makes the exponential moment finite. Condition the periodic Gibbs law at x , apply their Lemma 4.1, integrate, and use generalized Holder with $r_y = \vartheta |J_{xy}^\Lambda|/\kappa$ and $\sum_y r_y < 1$. This gives

$$e^{n_x} \leq e^C \prod_y e^{r_y n_y}, \quad n_x = \log \mu_{\Lambda, h}^{\text{per}}(e^{A_x}), \quad (8.2)$$

and hence

$$n_x \leq C + \frac{\vartheta}{\kappa} \sum_y |J_{xy}^\Lambda| n_y. \quad (8.3)$$

Here $\widehat{J}_\Lambda \leq 6c$. Periodic translation invariance and $\vartheta \widehat{J}_\Lambda < \kappa$ give $n_x \leq C/(1 - \vartheta \widehat{J}_\Lambda/\kappa)$. The compact-source uniformity of C is inherited from the stability constants checked in EXP-000781. For $0 < r \leq \beta$,

$$|\omega(0)|^2 \leq 2r^{-1} \|\omega\|_2^2 + 2r^{2\sigma} [\omega]_\sigma^2. \quad (8.4)$$

Given $a > 0$, choose $r \leq \beta$ with $2ar^{2\sigma} \leq \lambda_\sigma$ and then the freely available L^2 coefficient $\kappa \geq 2a/r$. The recursion and Holder control imply, at fixed β and on compact source intervals,

$$\sup_{\Lambda, h, x} \varphi_{\Lambda, h}(e^{a|q_x|^2}) < \infty \quad \text{for every } a > 0. \quad (8.5)$$

Choose $Q_L(q) = \eta(|q|/L)q$, equal to q on the L -ball and zero outside the $2L$ -ball, and truncate only the intersite coordinate. The truncated bond has norm $O(L^2)$. With $\rho = \rho_{\Lambda, h}$, its tail W_L and gradient over both bond endpoints and all components are polynomial factors on $\max(|q_x|, |q_y|) > L$, so

$$\|W_L\|_D^2 + \|[\log \rho, W_L]\|_D^2 \leq C_{a, N} L^N e^{-aL^2}. \quad (8.6)$$

To justify the identity, first impose a bounded outer cutoff $W_{L, M}$ on the common Schrodinger core and then send $M \rightarrow \infty$ by exponential domination. The exact Dirichlet identity is

$$\|[\log \rho, W_L]\|_D^2 = \frac{\beta \hbar^2}{\chi} \varphi(|\nabla_{(x, y)} W_L|^2). \quad (8.7)$$

If the missing structured comparison grows at most as $\text{poly}(L)e^{C_0 T L^2}$, choose $L_m = m^\alpha$, $0 < \alpha < 1/2$, and $a > C_0 T$. The static/modular tail and the bounded- interaction factorial error then both vanish. Constants depend on fixed β, a, h_0 and model data; no ground-limit uniformity is asserted. This scale arithmetic is conditional on the projected or multiplier estimate.

Primary source: Y. Kozitsky and T. Pasurek, <https://arxiv.org/abs/math-ph/0609045>.

9 Fixed-beta OS mixture precursor

For $0 < \lambda < 1$, assume the same reflection, underlying field space and positive-time test algebra. For $\mu_0 = \lambda\mu_+ + (1 - \lambda)\mu_-$, reflection positivity gives

$$q_0 = \lambda q_+ + (1 - \lambda)q_-, \quad N_0 = N_+ \cap N_-. \quad (9.1)$$

The map

$$[F]_0 \mapsto (\sqrt{\lambda}[F]_+, \sqrt{1 - \lambda}[F]_-)$$

is an isometry into the direct sum of the phasewise OS spaces. It need not be surjective. A common normal W-star interpretation additionally needs reconstruction functoriality on one formal field algebra. Central phase projections require disjointness; extremality helps only with a common-KMS- simplex theorem that implies it. The mixture μ_0 has not been proved to be the zero-source periodic finite-volume limit. This is only a fixed-beta envelope precursor, not a beta-independent Hamiltonian common alpha.

10 Devil's-advocate review

1. The star counterexample might be misread as nonexistence of the dynamics. This objection is upheld as a scope warning: equation (3.4) proves the same subflow is exactly unitary after all-order resummation.
2. The tree identity might be copied directly to the cubic lattice. This is dismissed by the square, where a second endpoint path remains after the chosen backbone is removed.
3. Static Gibbs tails might be assumed stable under every bounded evolved observable. This is dismissed by equation (7.1); modular half-strip or direct projected control is essential.

4. The OS mixture might be called the common Hamiltonian dynamics. This is dismissed: it is fixed-beta, its image need not be the full direct sum, and the Hamiltonian, central-decomposition and zero-temperature gates remain open.

11 Verification and route verdict

The primary implementation uses symbolic algebra and exact graph enumeration. The independent implementation uses only standard-library fractions, polynomial dictionaries and graph search. The integrated verifier requires fresh equality with the stored artefacts and checks all authority, scope, negative-result and PDF links.

The historical v1.1 first-passage and fifth-energy gate identifiers remain in the registry. Their active successors are the all-bond unitary Trotter gate and the projected-Duhamel modular- C^1 multiplier gate. Only after one yields a phase- and beta-independent common dynamics may the programme proceed to common-alpha KMS identification, algebraic ground selection, a broken-sector GNS gap, enlarged-counterterm continuum removal, and a same-Hamiltonian physical-empty comparison.

12 Reproduction

```
E:\Dev\TECT.venv\Scripts\python.exe -X utf8
codes\foundations\pre_a_cp1_st8_q3lock_modular_cutoff_unitary_resummation_
route_split.py
```

```
E:\Dev\TECT.venv\Scripts\python.exe -X utf8
codes\foundations\pre_a_cp1_st8_q3lock_modular_cutoff_unitary_resummation_
route_split_independent.py
```

```
E:\Dev\TECT.venv\Scripts\python.exe -X utf8
codes\foundations\pre_a_cp1_st8_q3lock_modular_cutoff_unitary_resummation_
route_split_verify.py
```

13 Result footer

Result ID:	PA-CP1-ST8-Q3LOCK-SECOND-WEIGHTED-ENERGY-MOMENT-AND-COMMON-ALPHA-CAUCHY-GATE-SPLIT.
Reusable result:	R-167 v1.2; no new result number.
Exploration:	EXP-000798.
Precise statement:	Fixed-s star/repeat first-passage target rejected; all-order star and unique-path tree repaired. Modular-mean, structured multiplier, and coordinate-cutoff static/modular tail theorems proved.
Scope:	T0 exact finite-volume and fixed-beta theorems, counterexamples, and conditional locality reductions.
Dependencies:	EXP-000781, EXP-000790, EXP-000794--EXP-000798; R-167 v1.0--v1.1.
Evidence grade:	ANALYTIC + EXACT EXECUTED + INDEPENDENT + PDF QA.
Reproduction command:	Run the three commands under Reproduction in the listed order, with the integrated verifier last.
Expected output:	Primary PASS 91/91; independent PASS 147/147; integrated PASS 288/288.
Falsification gate:	Any CCR sign, graph threshold, all-order phase, tree/square incidence, modular-mean constant, cutoff tail, hash, scope or provenance mismatch.
Tier before / after:	C6 T1 / C6 T1; no claim or tier change.
No-overclaim statement:	Trotter graph locality, projected Duhamel and structured modular-multiplier locality, common alpha/KMS, algebraic ground states, GNS gap,

regulator removal, continuum, physical empty and the
below-empty sign, functional selection, C6, CP1,
Sector A and Pre-A remain open.

Next required action: Prove either all-bond unitary Trotter graph-Lipschitz
closure or projected-Duhamel modular-C1 multiplier
locality in one preregistered faithful topology.